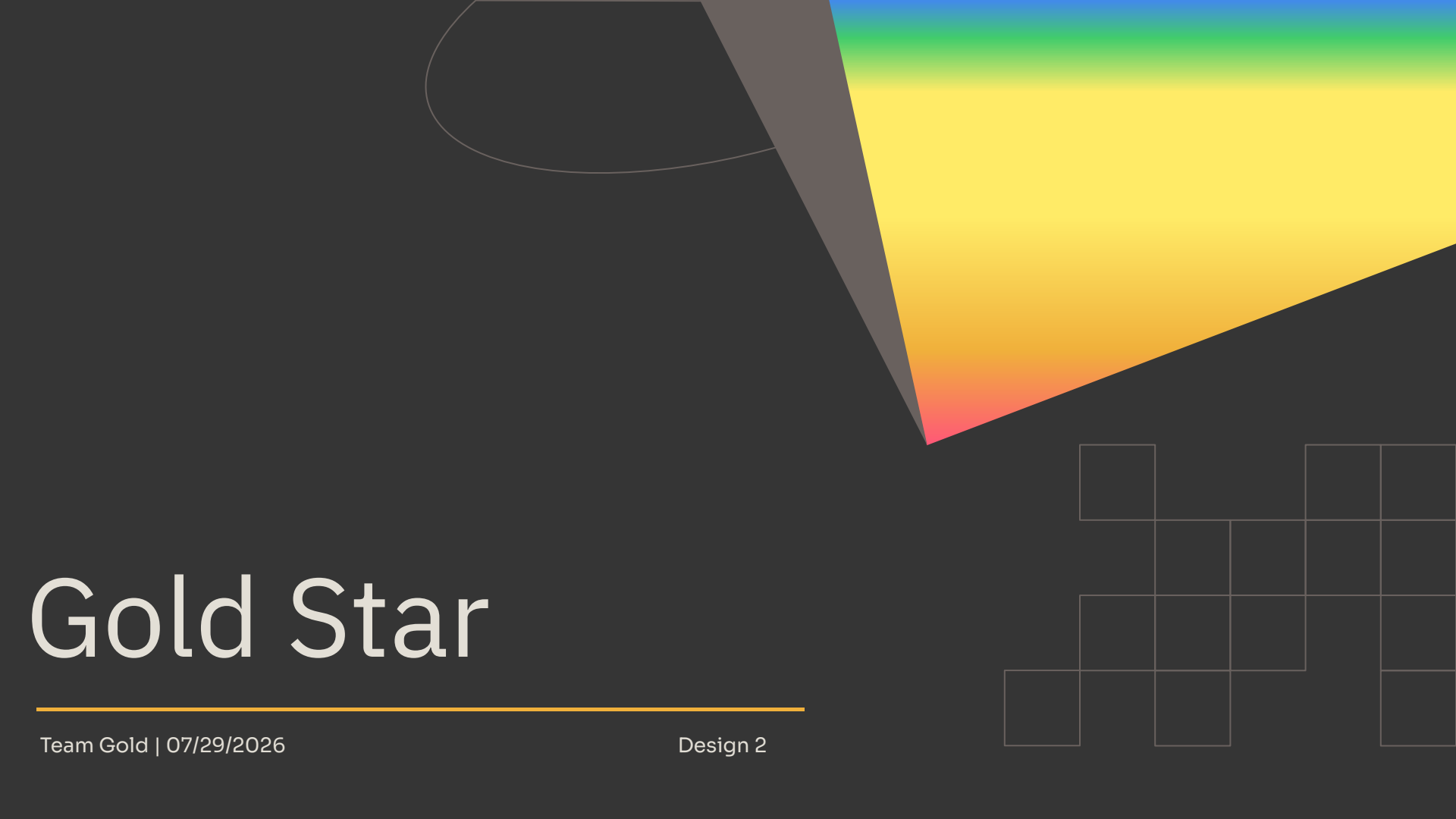




Gold Star

Team Gold | 07/29/2026

Design 2

Meet Team Gold



Matthew Allen



Aaron Brackett



Maggie Bragg



Jeffrey Bucher



Autumn Carey



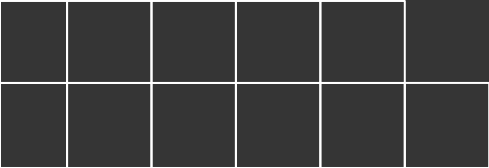
Evelyn Faye



Miguel Looney



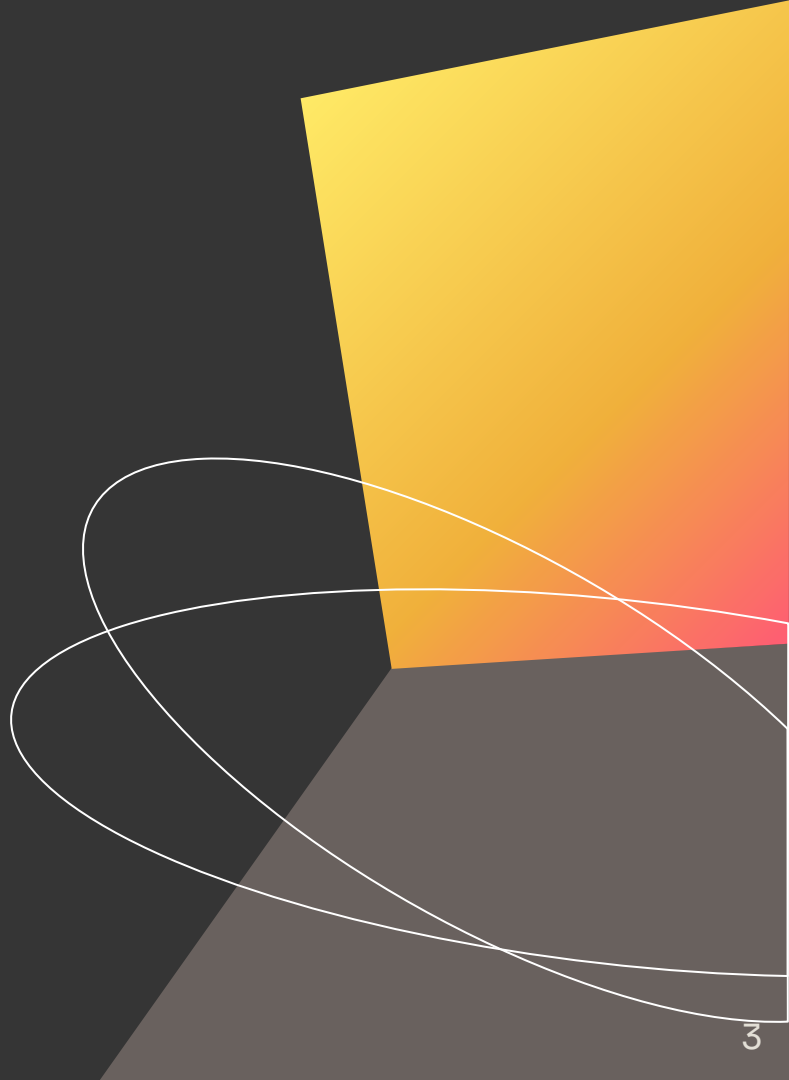
Lydia Nebuchadnezzar

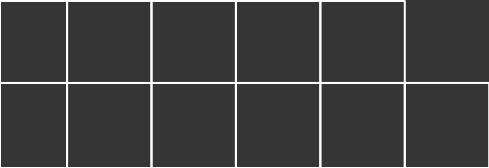


Elevator Pitch Problem

Millions of people struggle with ADHD symptoms — starting tasks, staying focused, and actually finishing what they begin.

While building routines is one of the most effective solutions, it's also one of the hardest things to do when consistency itself is the challenge.

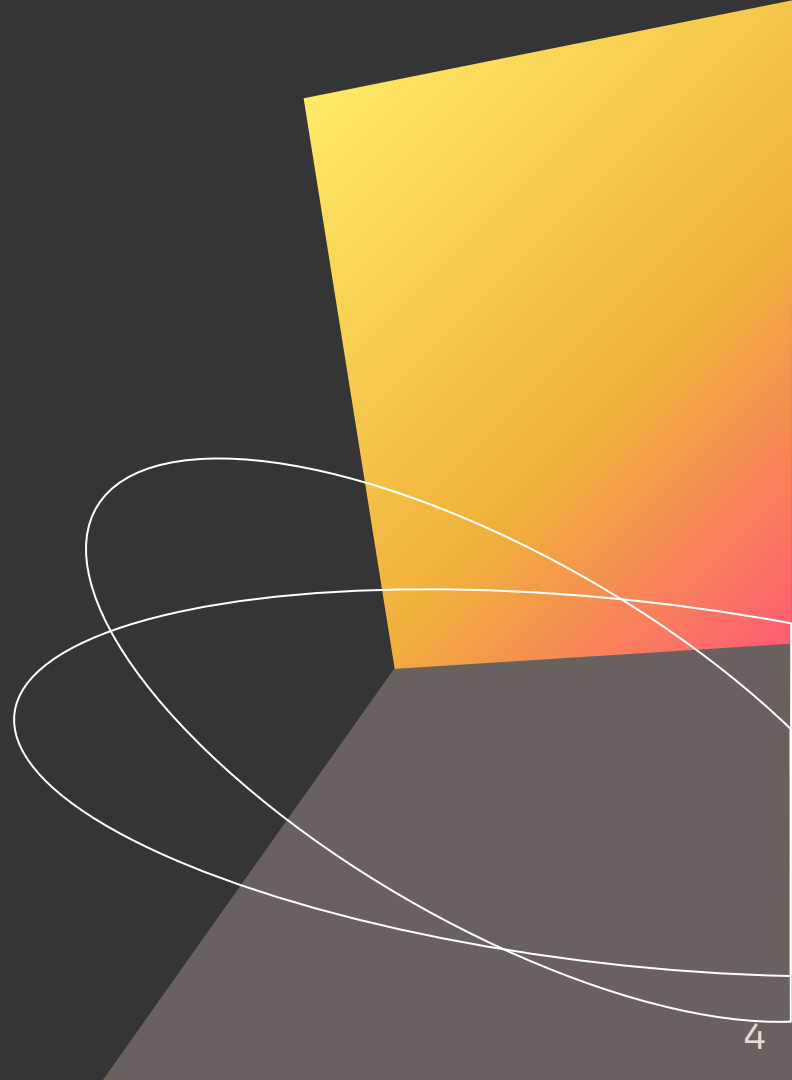




Elevator Pitch Solution

Gold Star is a web-based habit-building and time management platform designed specifically for individuals with ADHD.

It helps users organize their schedules, break down tasks, and build sustainable routines — making it easier to start, stay on track, and follow through everyday.



Contents

1. The Societal Problem
2. Our Solution
3. Development Tools
4. Major Functional Components
5. Risks
6. Work Breakdown Structure
7. Non-Trivial Algorithms
8. Database Schema
9. Real World Product vs Prototype
10. GUI Mockup

The Societal Problem

- Individuals with ADHD diagnoses or ADHD-like traits struggle with consistent routine development.
- ADHD is classified into three subtypes, or presentations, by the American Psychiatric Association
 - Inattentive
 - Hyperactive-Impulsive
 - Combined

The Societal Problem

- ADHD often persists beyond childhood.
 - As do the routine struggles that come with it!
- How long do routines last?
 - Anywhere from 3 days to “part of a month”
 - Until something happens—getting back into a routine after an interruption can feel extremely difficult for people with ADHD

The Societal Problem - Productivity-Altering Traits

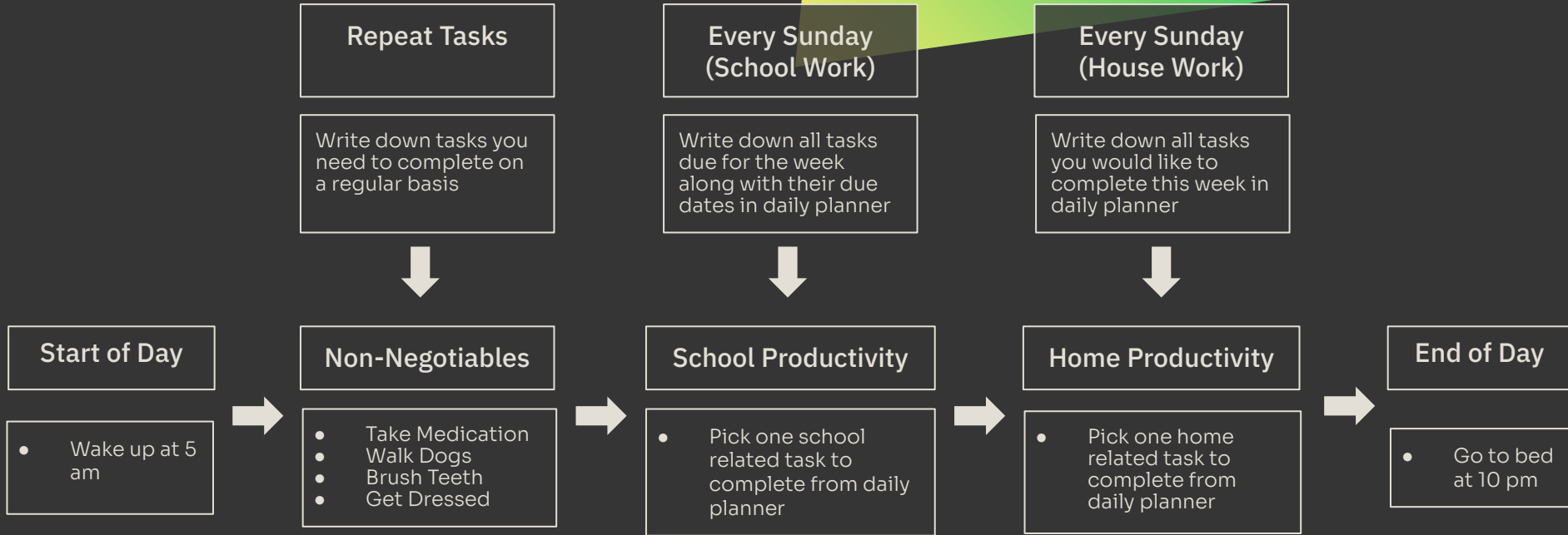
- ADHD may include certain productivity-altering traits shown below
 - Inattention
 - Hyperactivity
 - Impulsivity
 - Memory problems
 - Lack of focus

(Staley et al., 2026, p. 890; Advanced Psychiatry Associates, 2026)

The Societal Problem - Changing Routines

- Constantly changing or starting new routines can lead to:
 - Frustration
 - Anxiety
 - Avoidance
 - Reduced self-efficacy

Problem Flow - Before Gold Star



Automation & Partial Automation

- **Reminders**
 - Software responsible for reminding user
- **Adding Tasks**
 - User specifies tasks & timeframes, software retains tasks
- **Scheduling & Grouping Tasks**
 - User groups similar tasks by a task role, software fills free time with task roles

Repetition

- **Scheduling**
 - Schedules often last for multiple days or weeks
 - Prompt user to review schedule daily
 - Option to change schedule

Tedium

- **Creating New Schedules**

- Uncertain time slots may appear
- Higher odds of activity switching

- **Specifying Timeframes**

- How does our user know if a task's time frame is correct?

Subject To Error

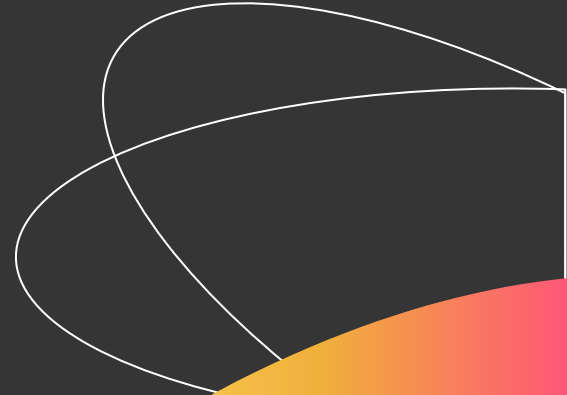
- **Forgetfulness**
 - User forgets to review their schedule
- **Task Deviations**
 - Tasks accomplished out of order
 - Tasks *not accomplished* at all

Solution

Gold Star is a web-based routine management platform designed to help individuals with ADHD organize their time, tasks, and daily responsibilities. It provides a flexible system that adapts to changing schedules and missed tasks while helping users build sustainable habits with structure and accountability.

Through an intuitive, calendar-based interface, users can plan, track, and maintain routines without feeling overwhelmed.

By combining flexibility with habit-building strategies, Gold Star helps users stay consistent, complete tasks, and better manage their daily lives.



Solution Flow - With Gold Star

KEY: ★ Changed with Gold Star
○ Original steps
★ Optional step

1. REPEAT TASKS

Input tasks that need to be completed regularly (e.g., “dishes”, “take out trash”)



Everyday Routine

Tasks that you do everyday.

★ 2. RIGID SCHEDULING

Input tasks that non-negotiable and need to be completed in specific time (e.g., “must be 6:00”)



Schedule Created For You

Your tasks are placed on the calendar.

★ 3. ADAPTIVE SCHEDULING

Input tasks that can be completed at any time within a timeframe (e.g., “between 7:00 to 10:00”)



Based on Productivity Level

Your schedule is adjusted based on your energy and productivity.

HOW IT WORKS

START OF DAY

- Wake up at 5 am
- Pick your productivity level
- Review and verify schedule



PUT IN YOUR SCHEDULE AUTOMATICALLY

- Take Medication
- Walk Dogs
- Brush Teeth
- Get Dressed



PUT IN YOUR SCHEDULE AROUND NON-NEGOTIABLES

Your fixed-time tasks are already scheduled.



SCHEDULE IS DETERMINED BY PRODUCTIVITY LEVEL

Tasks are prioritized based on how you feel and what needs to get done.



END OF DAY

- Go to bed at 10 pm
- Input your mood for the day
- Review schedule, rollover incomplete tasks

FLEXIBLE SCHEDULE CHANGE

At any point throughout your day, change the schedule how you see fit.

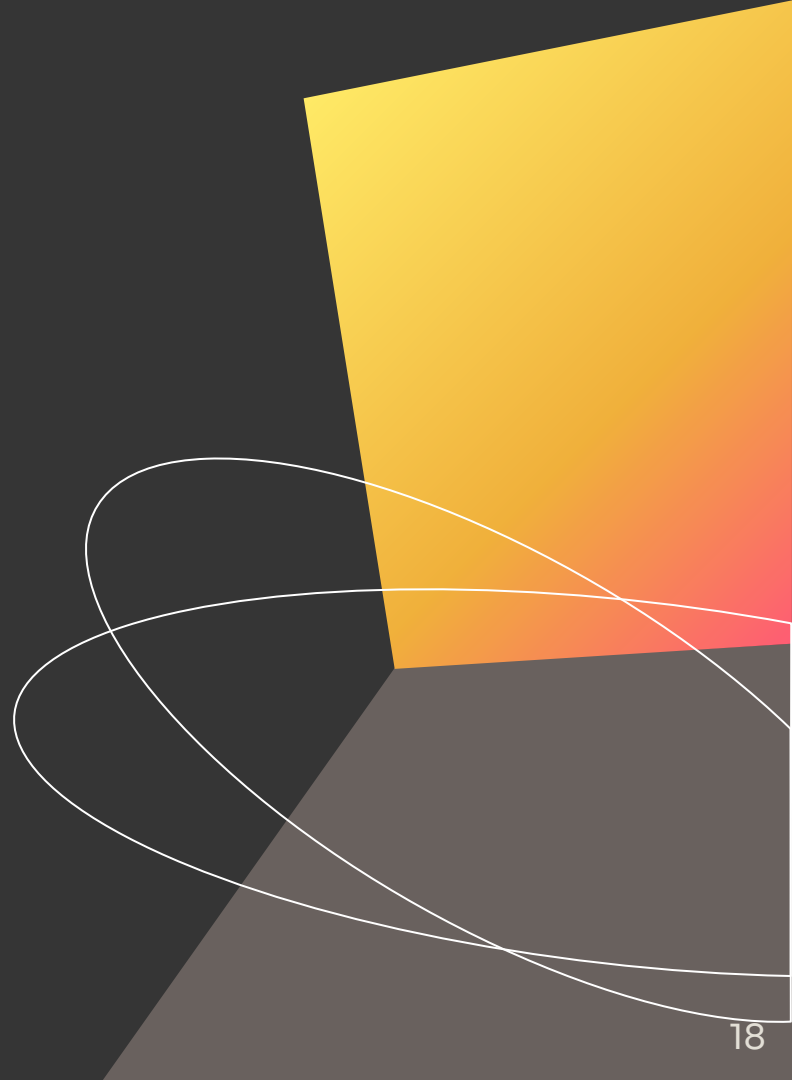
What It Will Do

- Create and manage routines
- Schedule daily tasks
- Track task completion
- Adjust routines for missed tasks
- Provide reminders and notifications
- Help users build long-term habits



What It Will *Not* Do

- Diagnose ADHD or any medical condition
- Replace therapists, coaching, or medical treatment
- Provide medical advice or clinical guidance
- Guarantee productivity or eliminate all missed tasks
- Manage financial accounts, banking or healthcare records
- Function as an emergency or crisis support tool



Competition Matrix

Feature :	Gold Star	Google Calendar	Habitica	Todoist
Calendar View	✓	✓	✗	✓
Adaptive Scheduling	✓	✗	✗	✗
Task Prioritization	✓	✗	✓	✓
Habit Stacking	✓	✗	✓	✗
Reward System	✓	✗	✓	✗

Development Tools

Development Type	Frontend	Backend
IDE	Visual Studio Code	Visual Studio Code
Version Control	Git & GitHub	Git & GitHub
Continuous Integration & Deployment	GitHub Actions & Workflows	GitHub Actions & Workflows
Programming Language	HTML, CSS, & JavaScript	Node.js
Testing Framework	Jest (tentative)	Jest (tentative)
Documentation Tool	Javadoc (tentative)	Javadoc (tentative)

Major Functional Components

Presentation Layer

User Interface (WebApp)

- Calendar View
- Routine Dashboard
- Productivity Level
- Statistic Dashboard

Application Layer

Webserver

- Node.js
- Registration, Login,
Session

Notification Handling

- WebSockets

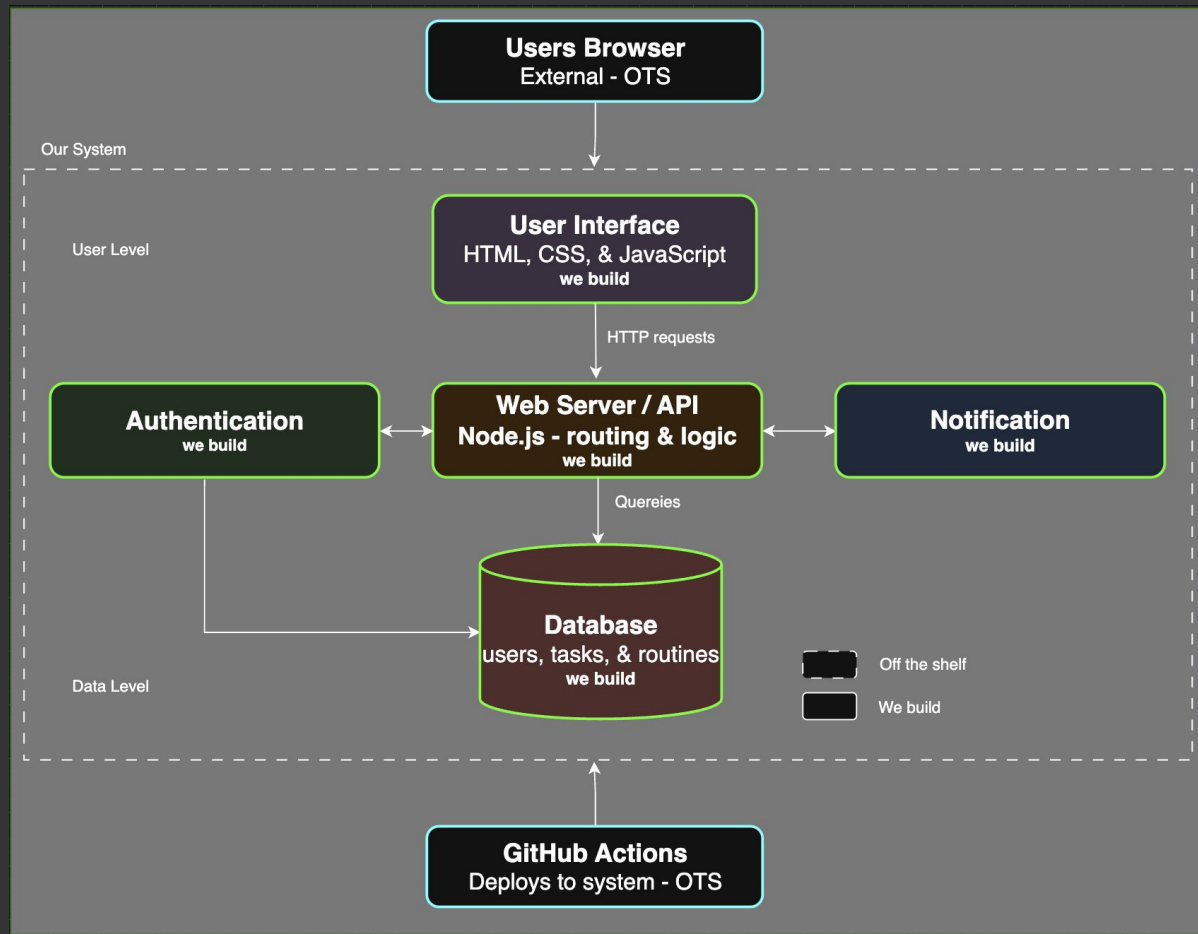
Data Layer

PostgreSQL

- Encrypted
- User Data
- Routine data

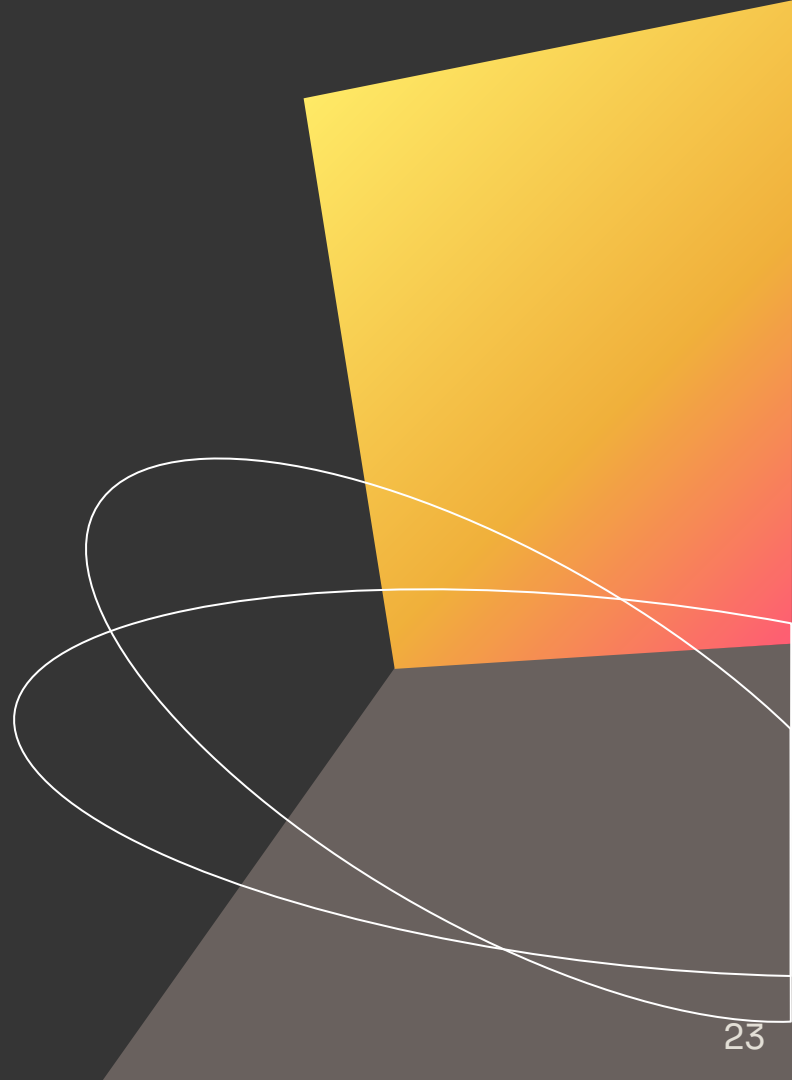
Integration APIs

MFC Diagram



Risks

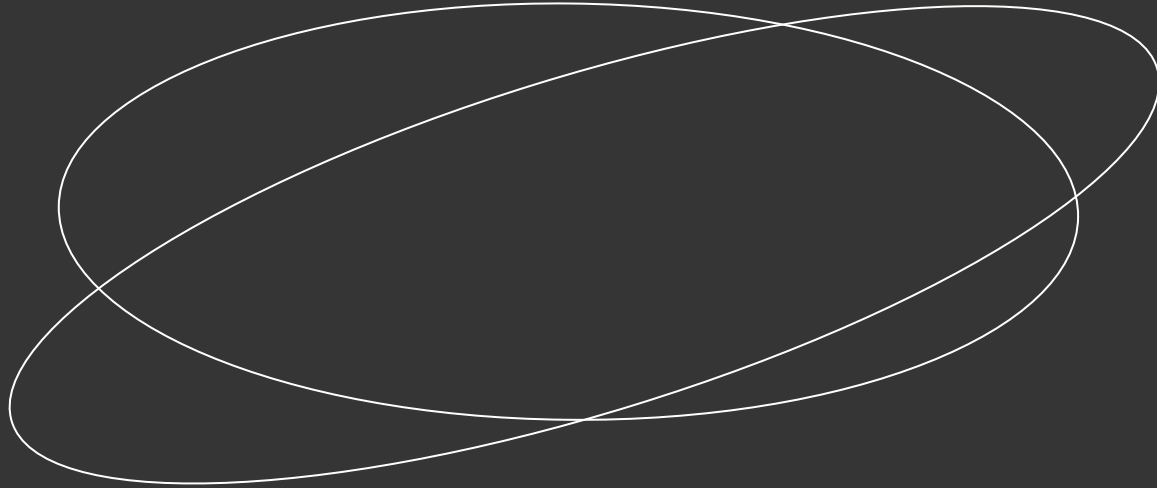
1. Customer
2. Legal
3. Security
4. Technical



Customer Risks

Building for ADHD means designing *against* abandonment

1. Attention
2. Emotion
3. Consistency
4. Uncertainty
5. “One Size Fits None!”
6. Planning ≠ Doing



Customer Risks

Attention

- Users will forget to use the application
- Users will feel overwhelmed by the application
 - Total amount of features
 - Complexity of features

Risk Categories	Before Mitigation
Probability	5
Impact	5

Customer Risks

Attention Mitigation

- Tool Tips
 - Brief tips on features
 - Incentivizes user to try features at their pace
- “Gradual Mode” Option
 - Introduce program features gradually as consistent user activity is formed
 - Prompt user feature availability option after registration
 - Gradually introduce features
 - Or have immediate access to all features upon login

Risk Categories	After Mitigation
Probability	3
Impact	5

Customer Risks

Emotion

- Missing tasks & routines feels *devastating!*
 - Can bring shame, not motivation
 - Can make the application feel “high-stakes”

Risk Categories	Before Mitigation
Probability	5
Impact	5

Customer Risks

Emotion Mitigation

- Accommodating UI
 - Do not let the user feel ashamed!
 - No in-program consequences for missing tasks
 - No red text or other negative imagery for missing tasks **by default**
 - Allow UI customization: what seems discouraging to 1 user is not universal to all users
- “Task Importance” Feature
 - User can group tasks based off urgency

Risk Categories	After Mitigation
Probability	3
Impact	5

Customer Risks

Consistency

- Gold Star can be a novelty for a while
- We do not have a second chance to draw users back

Risk Categories	Before Mitigation
Probability	4
Impact	4

Customer Risks

Consistency Mitigation

- Reward System
 - Offer virtual “Gold Stars” when completing tasks and schedules
 - No FOMO* mechanics, these can alienate users.
- Milestones
 - Celebrate sentimental accomplishments
 - Task/schedule completions
 - Birthdays
 - Graduations
 - Make users feel special!
 - Make users feel attached!

Risk Categories	After Mitigation
Probability	3
Impact	4

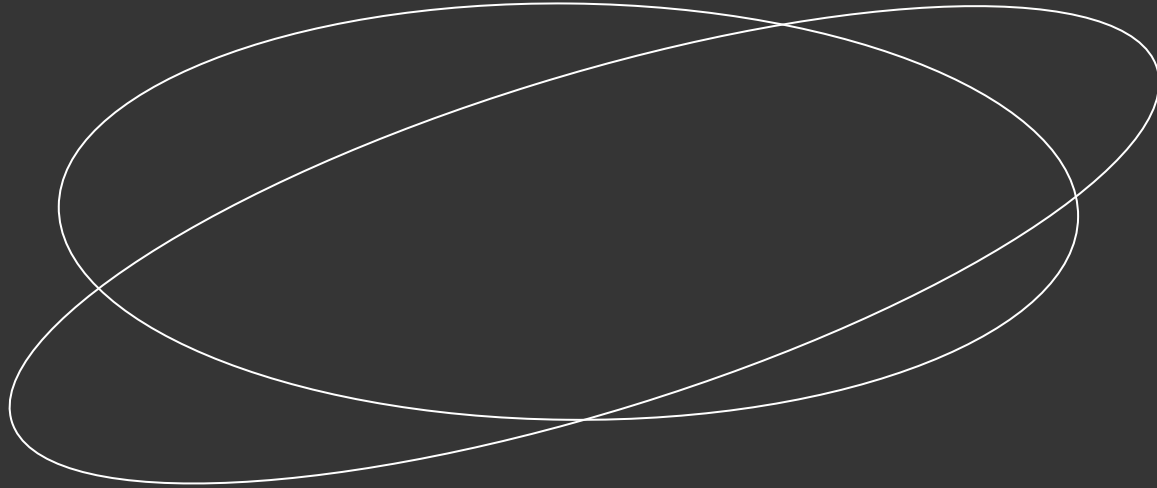
*(Fear of Missing Out)

Customer Risks

Uncertainty

- Life is messy
 - Real life does not always follow a schedule
 - App must bend without breaking

Risk Categories	Before Mitigation
Probability	5
Impact	4



Customer Risks

Uncertainty Mitigation

- Adaptive Scheduling
 - Allow user to adapt their schedule at any time during their day
- No Perfectionist Records
 - Avoid gamification of daily or weekly streaks
- “Rollover” System
 - Allow unfinished tasks at the end of a day to “rollover” into the next
 - Rolling over tasks is up to the user’s discretion
 - Gives users agency
 - Avoids pressure if excessive task rollover seems stressful

Risk Categories	After Mitigation
Probability	3
Impact	4

Customer Risks

“One Size Fits None!”

- ADHD looks different to everyone
 - Rigidity may feel alienating
 - Some may embrace it
 - Others may feel *afraid* of it

Risk Categories	Before Mitigation
Probability	4
Impact	4

Customer Risks

“One Size Fits None!” Mitigation

- Time Frames
 - Not all events require a rigid schedule
 - Some may only need estimated ranges
- Mood and Productivity Level Settings
 - User is prompted to toggle daily productivity and mood levels
 - Intended as 2 separate levels
 - Example: Not all happy days guarantee high-productivity
- Adaptive Scheduling
 - Change routines at any time in response to unexpected changes

Risk Categories	After Mitigation
Probability	3
Impact	4

Customer Risks

Planning ≠ Doing

- Scheduling may be easy
 - Following through is a different story
 - Often where ADHD individuals are stuck

Risk Categories	Before Mitigation
Probability	5
Impact	4

Customer Risks

Planning ≠ Doing Mitigation

- Habit Stacking
 - User mixes new & solidified habits
 - Avoids task overload
- Alternating Activities
 - Going from a hard task to an easy task
 - Tasks can be changed via Adaptive Scheduling
 - Avoids burnout
- Pomodoro Technique
 - Breaks dispersed throughout productivity
 - User specifies break lengths & disperse times
 - System reminds user
 - Avoids mental fatigue

Risk Categories	After Mitigation
Probability	3
Impact	4

Legal Risks

Building for ADHD involves *setting boundaries* and *meeting guidelines*

1. Medical Liability
2. Website Accessibility

Legal Risks

Medical Liability

- Users may assume Gold Star substitutes:
 - Licensed therapists
 - Medical professionals
 - Clinical treatments

Risk Categories	Before Mitigation
Probability	3
Impact	5

Legal Risks

Medical Liability Mitigation

- Legal Disclaimer
 - Specify Gold Star's intent as a productivity tool only
 - It is not advised to replace therapists, healthcare providers, or medical procedures with this program

Risk Categories	After Mitigation
Probability	2
Impact	5

Legal Risks

Website Accessibility

- Gold Star is a web program
- Web programs must follow accessibility standards set by:
 - Americans with Disabilities Act (ADA)
 - W3C Web Accessibility Initiative (WAI)

Risk Categories	Before Mitigation
Probability	3
Impact	5

Legal Risks

Website Accessibility Mitigation

- Perceivability
 - Ensure all web material can be distinguished visually and aurally
 - Adaptive screen layouts
 - Screen reading support (via Semantic HTML)
 - Avoiding imperceptible contrast and color levels
- Operability & Stability
 - Implement WAI-satisfactory navigational standards
 - Ensuring all web pages go through HTML validation

Risk Categories	After Mitigation
Probability	2
Impact	5

Security Risks

Building for everyone involves *building safety* for everyone

1. Compromised Accounts
2. SQL Injection

Security Risks

Compromised Accounts

- Gold Star implements a user account system
 - Passwords will be retrieved in a data breach
 - Unauthorized account access
 - Users have tendencies to *reuse* passwords
 - Unauthorized access *elsewhere*

Risk Categories	Before Mitigation
Probability	4
Impact	5

Security Risks

Compromised Accounts Mitigation

- Two-Factor Authentication
 - Required for all Gold Star accounts
- Password Salting
 - Passwords are assigned random, uniquely encrypted strings in account database
 - Passwords used by more than 1 account have unique encryptions
- Passphrase Generation
 - Generates 4 random dictionary words together as a password
 - Recommended for users who want ideas for safer passwords

Risk Categories	After Mitigation
Probability	2
Impact	5

Security Risks

SQL Injection

- Gold Star is a web program dependent on database software
 - Web forms reliant on SQL queries can be overridden
 - Overridden queries lead to unauthorized database access

Risk Categories	Before Mitigation
Probability	4
Impact	5

Security Risks

SQL Injection Mitigation

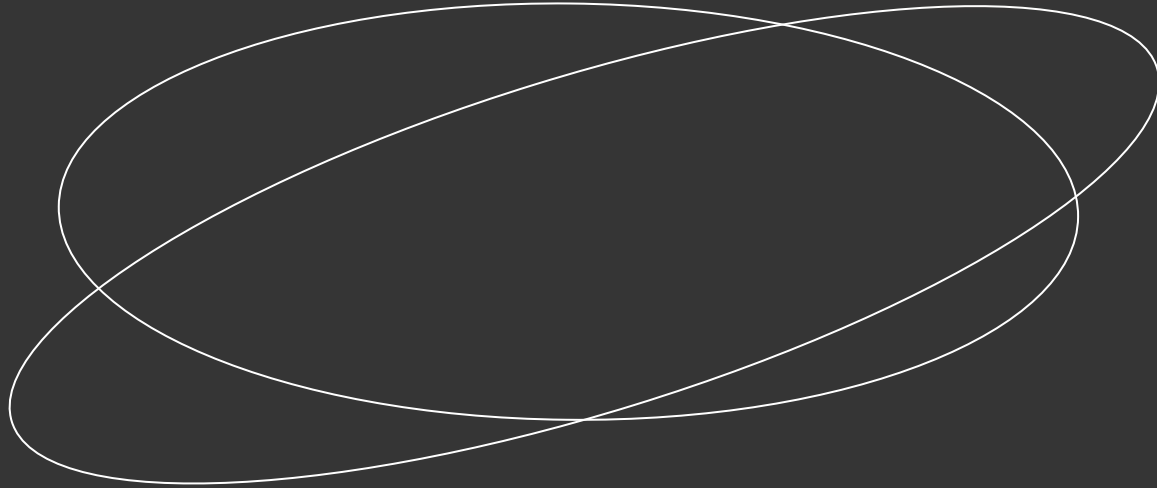
- Prepared Statements
 - Keeps SQL code and user's input values separate
 - Ensures input values will not override queries
- Server-Side Validation
 - Client-side validation can be bypassed, server-side cannot
 - Results in mandatory Prepared Statements
 - Verifies all input rules are followed

Risk Categories	After Mitigation
Probability	2
Impact	5

Technical Risks

Building for everyone involves *project stability* and *accuracy*

1. Database Unavailability
2. General Software Bugs



Technical Risks

Database Unavailability

- Gold Star is dependent on database software
 - Databases will not always be available
 - Can affect user accounts
 - Can affect changing routines

Risk Categories	Before Mitigation
Probability	3
Impact	5

Technical Risks

Database Unavailability Mitigation

- Backup Databases
 - Ensure at least one backup database is available
 - Ensure the backup database is routinely updated
 - Backup's data should be as recent as possible compared to the original
 - When backups are used mid-downtime, copy all changes from backup to original

Risk Categories	After Mitigation
Probability	3
Impact	4

Technical Risks

General Software Bugs

- Software is prone to errors, faults, and failures
 - Will lead to inaccurate actions
 - Leads to user confusion and possible stress

Risk Categories	Before Mitigation
Probability	3
Impact	5

Technical Risks

General Software Bugs Mitigation

- Test-First Development
 - Ensures enough code is written to justify a new unit test
 - All features guaranteed to run through verification for correctness
 - Guarantees mandatory regression testing for feature edits
- Integration and System Testing
 - Avoids only testing in an isolated perspective
 - Ensures features and functions are working together correctly

Risk Categories	After Mitigation
Probability	3
Impact	3

Work Breakdown Structure

Frontend (UI)	Backend (API)	Database (PostgreSQL)	Algorithms	Testing & Deployment
Homepage	Authentication	Users	Task Prioritization	Unit Testing
Dashboard	API Routes	Tasks	Adaptive Scheduling	Integration Testing
Calendar	Reward Logic	Habits	Rollover System	System Testing
Task Manager	Notifications	Rewards	Habit Stacking	Bug Fixes
Statistics	Progress Management	User Progress	Task Grouping	Deployment

Non-Trivial Algorithms

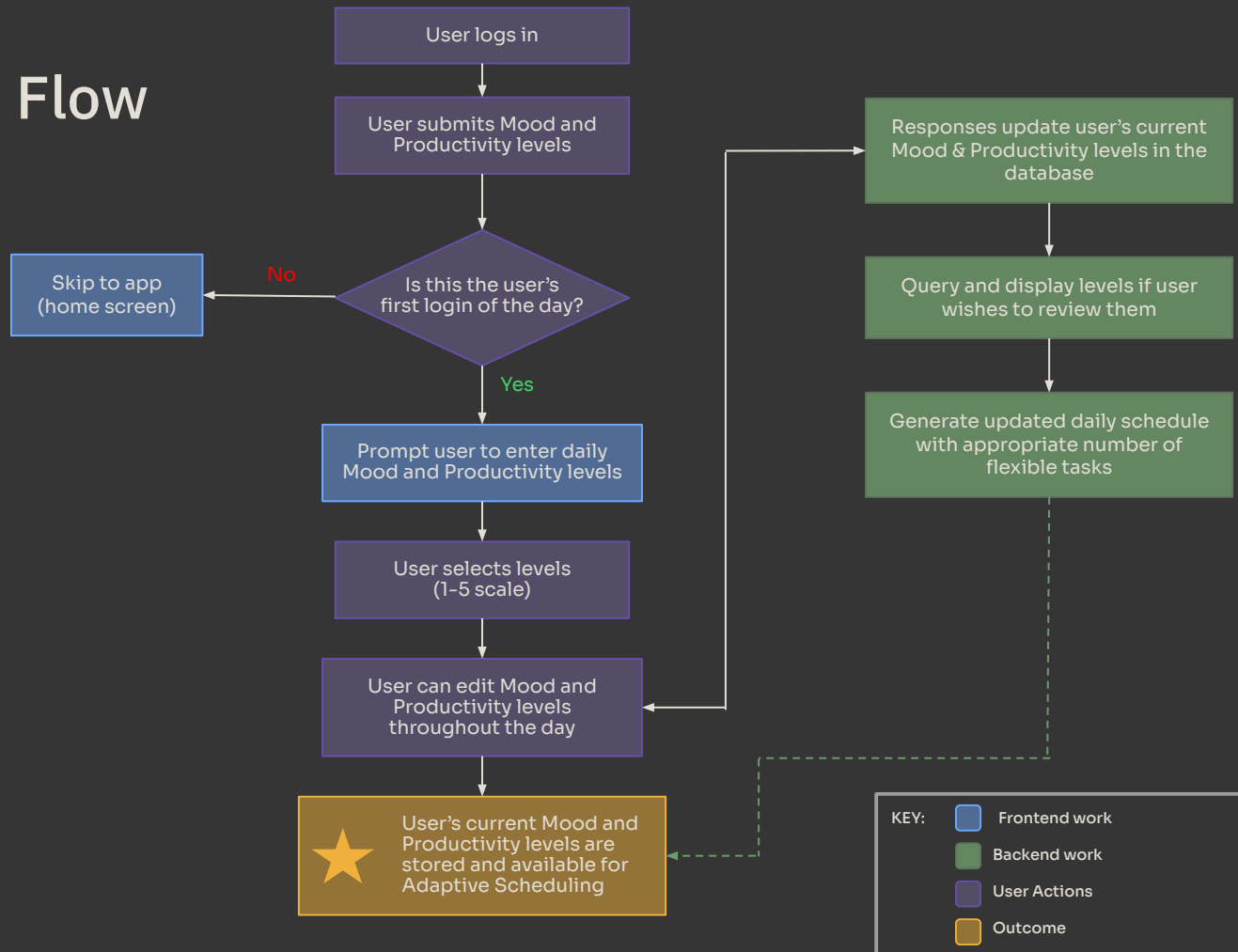
1. Productivity Determination
2. Adaptive Scheduling
3. Rigid Scheduling
4. Task Grouping
5. Task Prioritization
6. Rollover System

Non-Trivial Algorithms

Productivity Determination

- Details
 - User must enter daily Mood and Productivity levels when first logging in for the day
- Frontend Work
 - User enters levels if they logged in for their first time daily
 - Mood and Productivity are scaled from 1 to 5
 - Mood Default: Represented with preset emojis
 - Productivity Default: Represented with preset phrases (“Highly focused”, “Not at all”, etc.)
 - Offer setting for numerical scaling as alternative representation
 - User can edit levels throughout their day
- Backend Work
 - Responses update user’s current Mood & Productivity levels in database
 - Query and display levels if user wishes to review them

Productivity Determination Flow

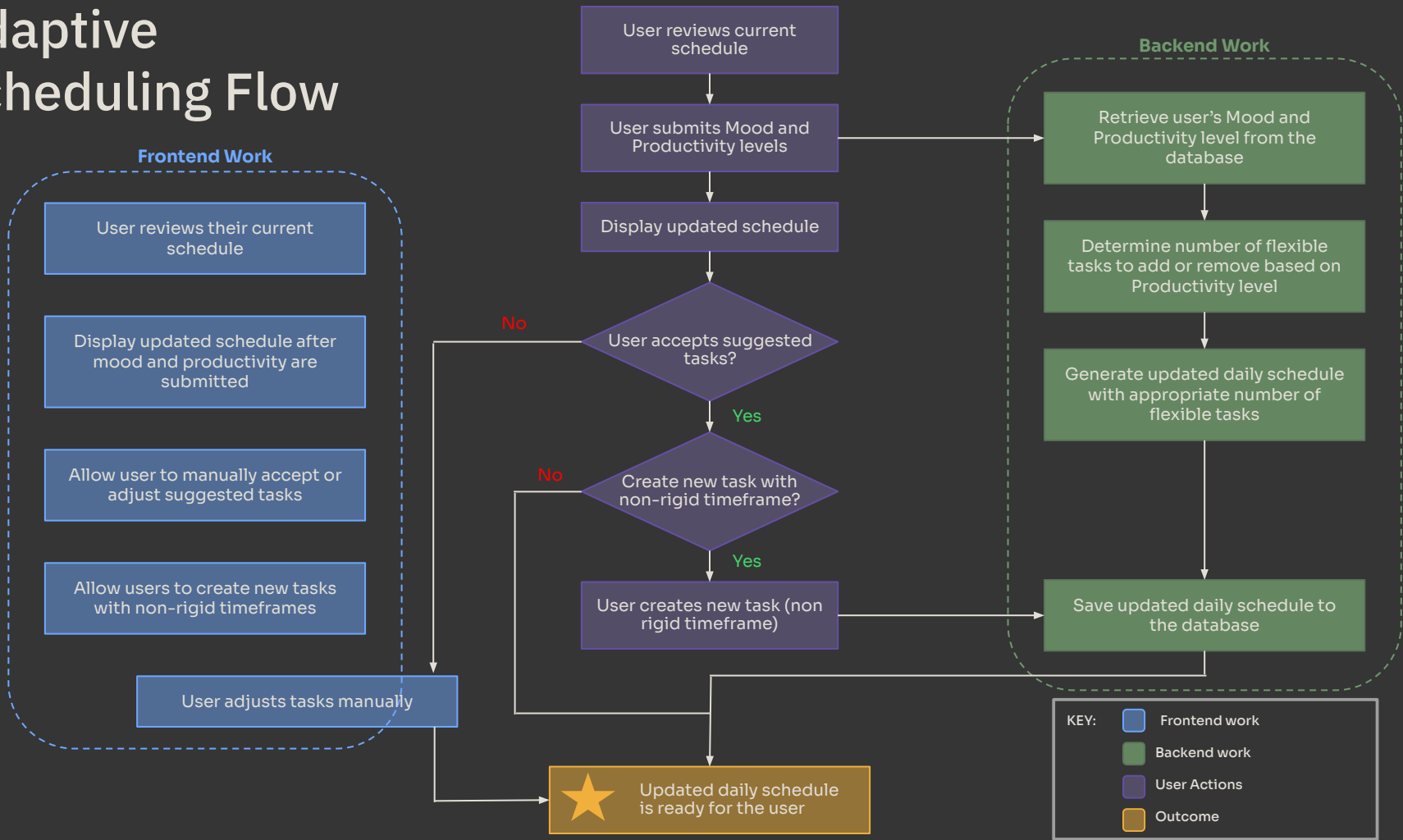


Non-Trivial Algorithms

Adaptive Scheduling

- Details
 - Distribution of tasks with non-rigid timeframes depends on the user's daily Mood and Productivity levels
 - Higher Productivity level = more non-rigid tasks
 - Lower Productivity level = less non-rigid tasks
- Frontend Work
 - User reviews their current schedule
 - Display updated schedule after mood and productivity are submitted
 - Allow user to manually accept or adjust suggested tasks
 - Allow user to create new tasks with non-rigid timeframes
- Backend Work
 - Store newly created tasks by user into the database (if applicable)
 - Retrieve user's Mood and Productivity level from the database
 - Determine number of flexible tasks to add or remove to daily schedule
 - Generate and save the updated daily schedule

Adaptive Scheduling Flow

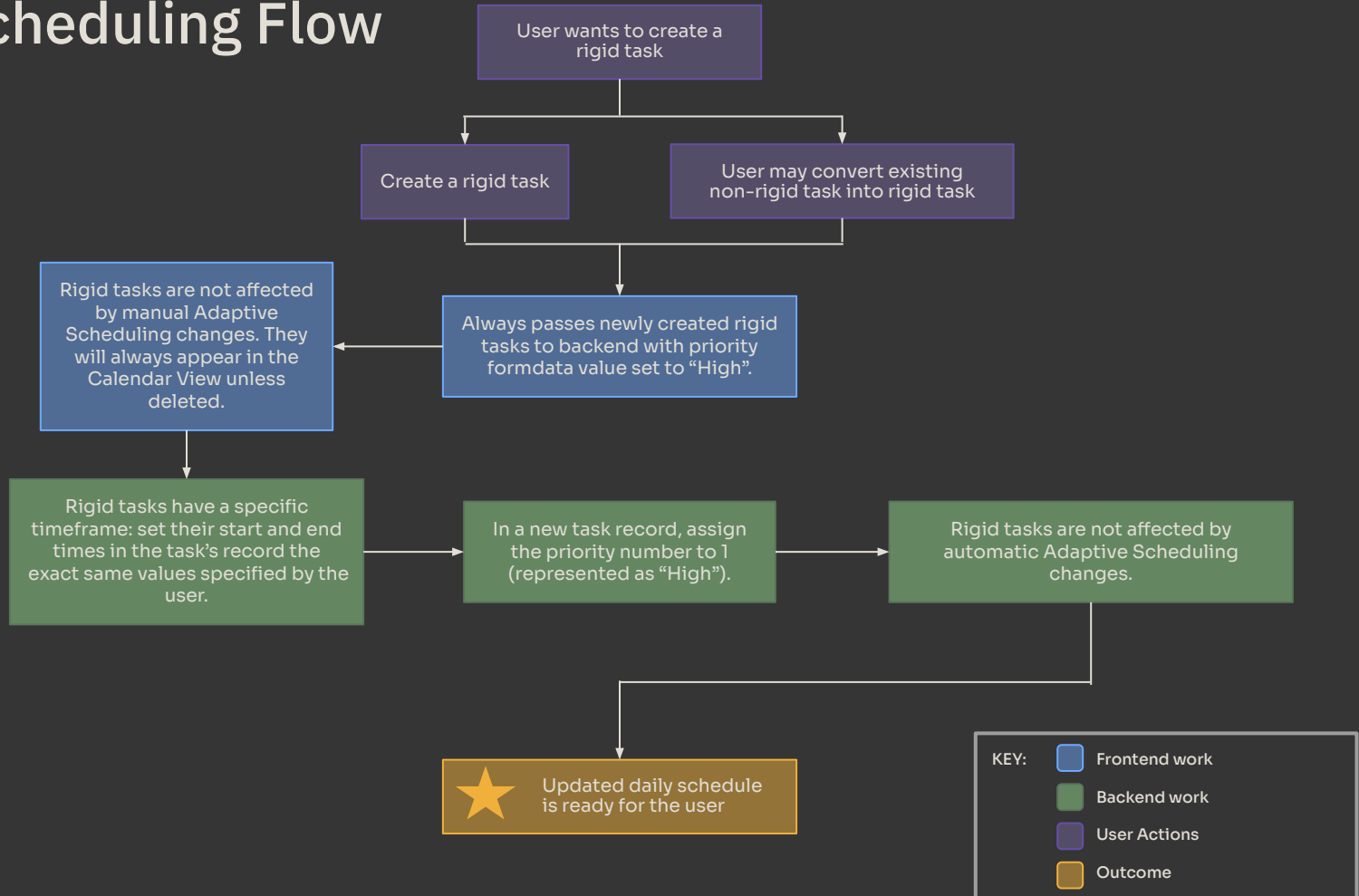


Non-Trivial Algorithms

Rigid Scheduling

- Details
 - Tasks that are given specific times by the user are placed into the daily schedule
 - Rigid tasks are unaffected by Adaptive Scheduling changes
 - Will always appear on schedule regardless of user's productivity level
- Frontend Work
 - Calendar View always displays all rigid tasks
 - After creating or editing rigid task, assign priority formdata value to "High"
 - Tasks with non-rigid timeframes can also be changed into a rigid task
- Backend Work
 - Store rigid task information into database
 - Query and display all rigid tasks from database when user is in Calendar View

Rigid Scheduling Flow

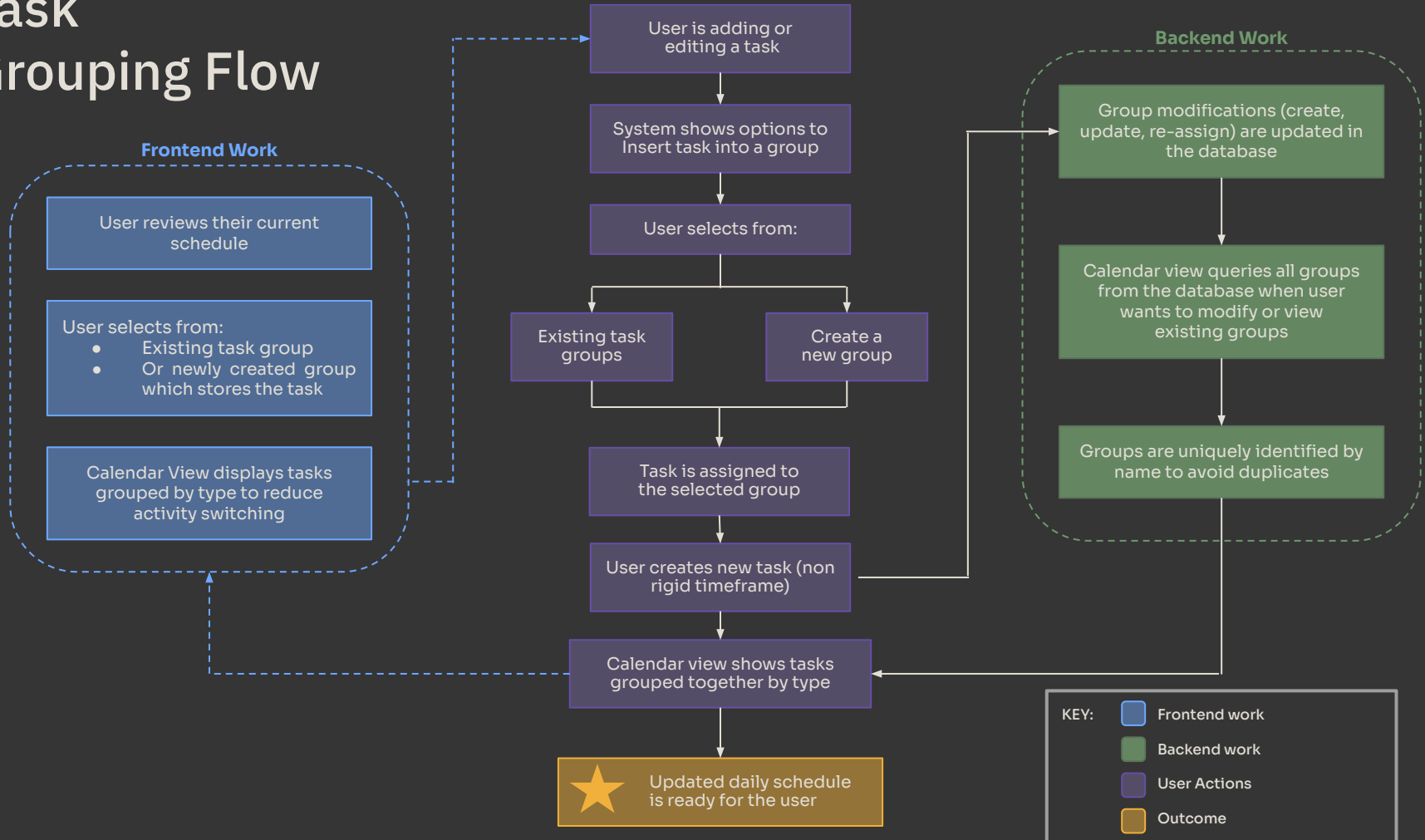


Non-Trivial Algorithms

Task Grouping

- Details
 - User groups tasks by type (cleaning, homework, appointments, etc.)
 - Adaptive Scheduling will ensure grouped tasks are assigned together
 - Reduces activity switching
- Frontend Work
 - User is currently adding or editing a task
 - Provide options to insert task into a group, or re-assign the task's group
 - User selects from:
 - Existing task groups
 - Or newly created group which stores the task
- Backend Work
 - Group modifications are updated in the database
 - Calendar View queries all groups when user wants to modify or view existing groups
 - Groups are uniquely identified by name to avoid duplicates

Task Grouping Flow

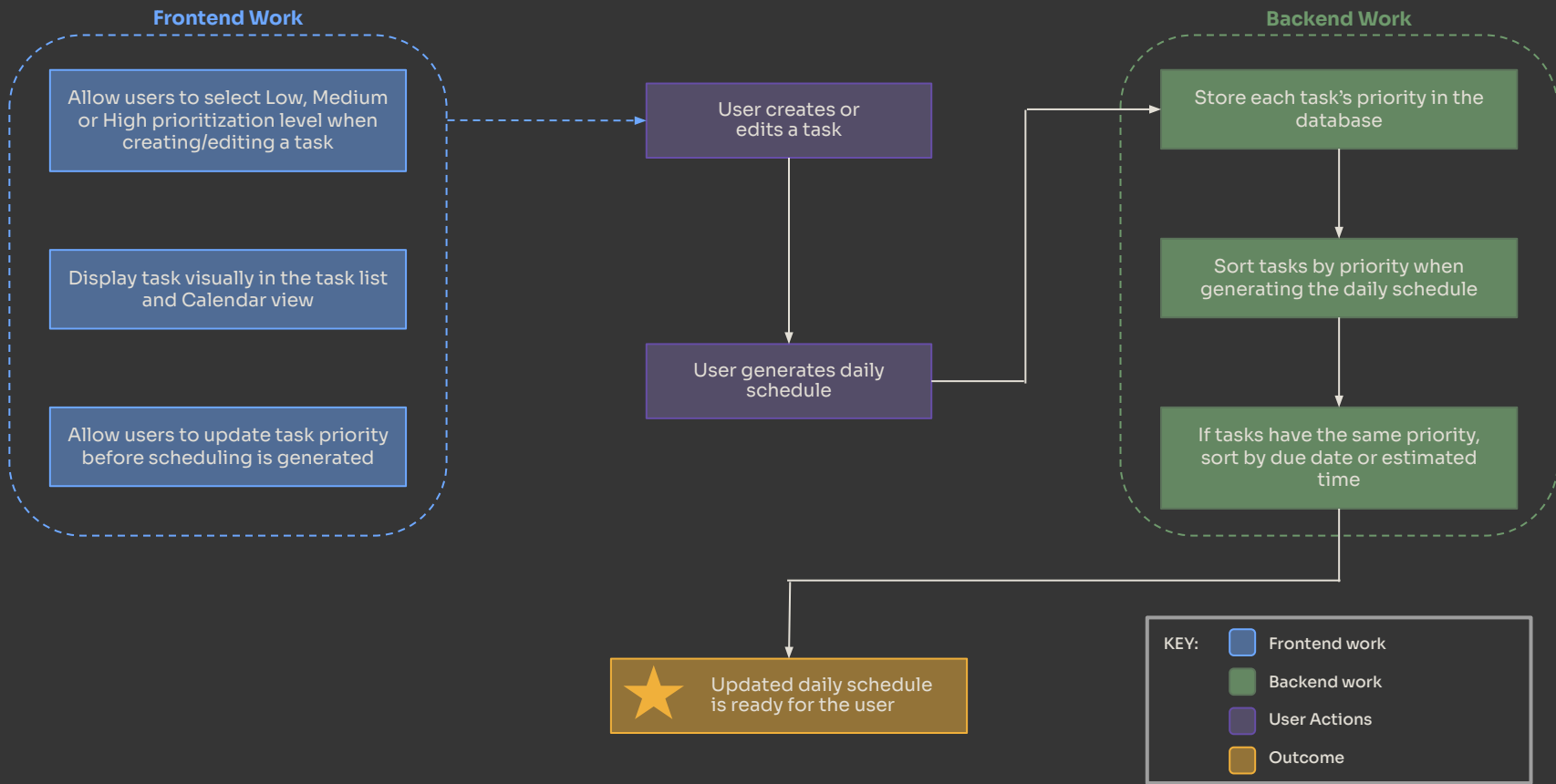


Non-Trivial Algorithms

Task Prioritization

- Details
 - User gives priority levels to tasks
 - Low, Medium, High
 - Adaptive Scheduling will ensure tasks are scheduled by priority level
- Frontend Work
 - Allow users to select Low, Medium, or High priority when creating/editing a task
 - Display task priority visually on the task list and Calendar View
 - Allow users to update task priority before the schedule is generated
- Backend Work
 - Store each task's priority in the database
 - Sort tasks by priority when generating the daily schedule
 - If tasks have the same priority, sort by due date or estimated time

Task Prioritization Flow

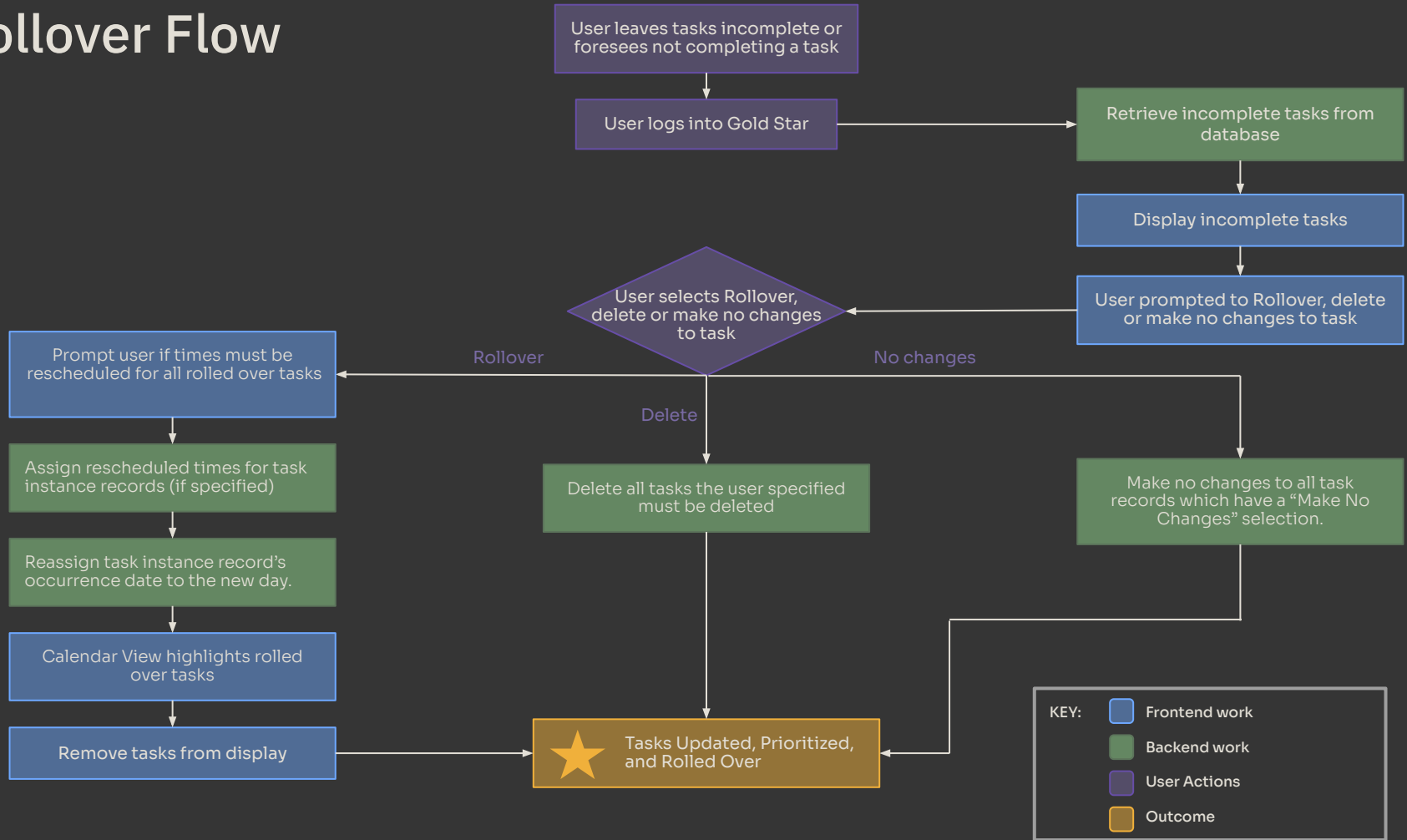


Non-Trivial Algorithms

Rollover System

- Details
 - If user has incomplete tasks or foreseen incomplete tasks for the day, missed tasks can be bumped up 1 level in priority and rolled over to the next day
- Frontend Work
 - Display incomplete tasks from previous day
 - Asks user to Rollover task (can decline)
 - Highlight rolled-over tasks in the daily schedule
- Backend Work
 - Retrieve incomplete task from database
 - Increase Rollover task priority (if applicable)
 - Update task date and save changes to the database

Rollover Flow



Database Schema

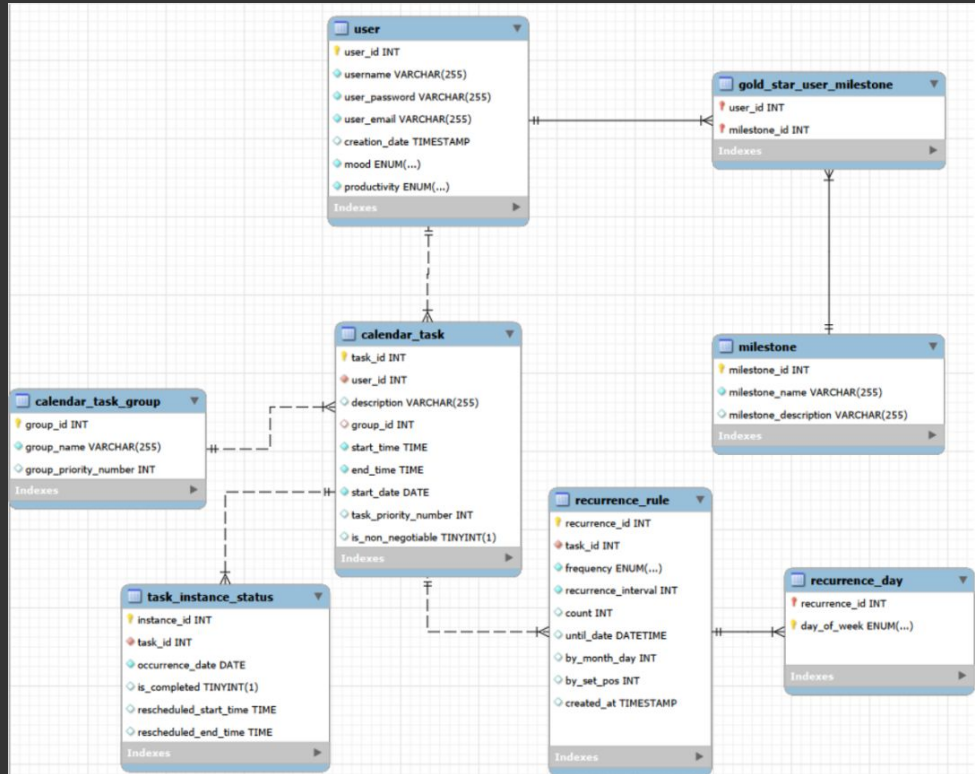


Diagram was made through MySQL Workbench

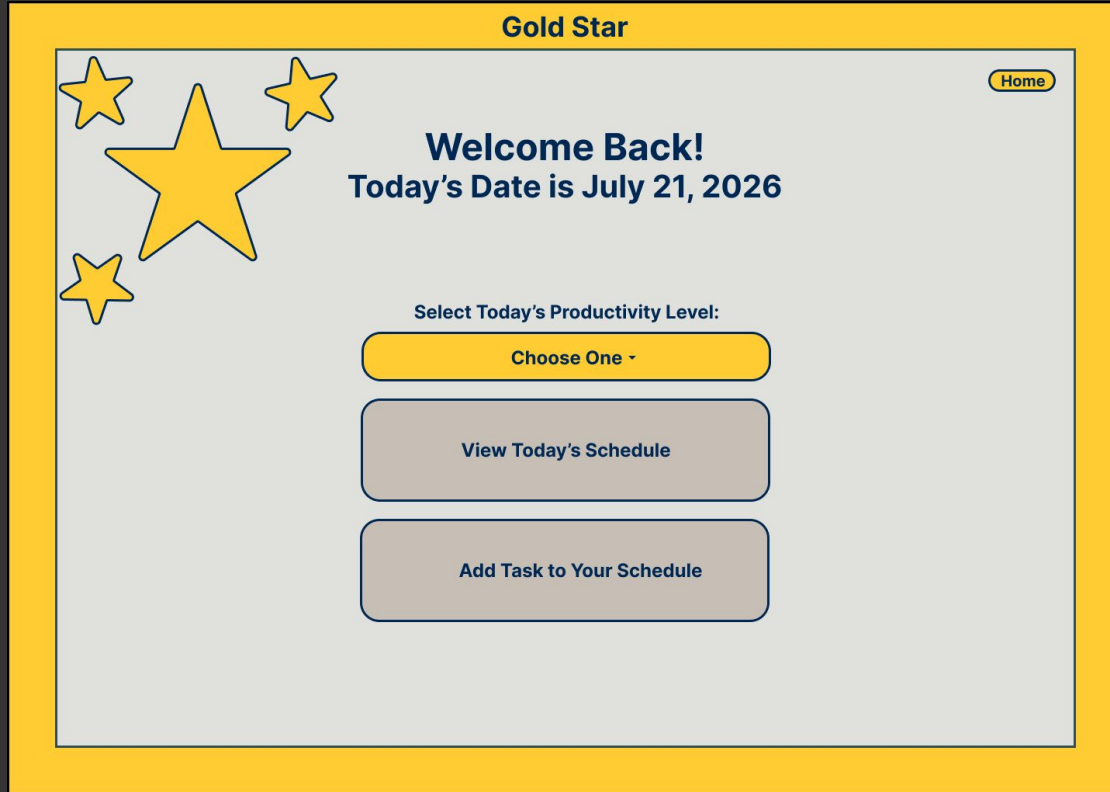
Real World Product vs Prototype

Features & Functionality	Real World Product	Prototype (Planned)
Calendar View	✓	✓
Adaptive Scheduling	✓	✓
Task Prioritization	✓	✓
Habit Stacking	✓	✓
Reward System	✓	✗
Customizable GUI	✓	✗

GUI Mockup

1. Welcome Page
2. Calendar View
3. Task Configuration

GUI Mockup - Welcome Page



GUI Mockup - Calendar View

Gold Star

Home

Today's Tasks: July 21, 2026

5:00 AM:	Take Medication / Get Dressed / Brush Teeth Walk Dogs
6:00 AM:	Breakfast Gym
7:00 AM:	CS410 Lab Outline
8:00 AM:	Break CS410 Lab Outline
9:00 AM:	Break Clean Bathroom
10:00 AM:	Break CS417 Homework 8
11:00 AM:	Break CS417 Homework 8
	Break

GUI Mockup - Task Configuration

Gold Star



Home

Add Task

Name:

Complete By:

July 22, 2026 ▾

Starts:

Anytime ▾

Ends:

Anytime ▾

Priority:

High ▾

Submit Task

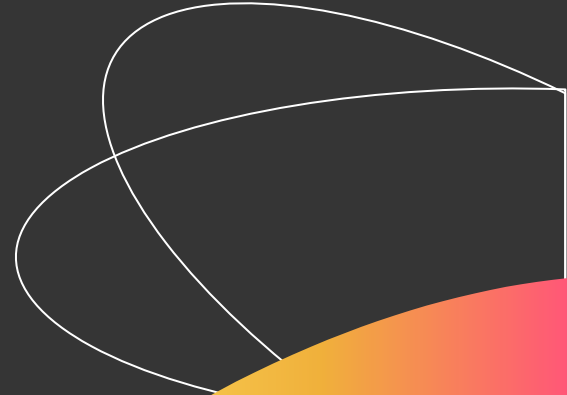
Conclusion

Millions of people struggle with ADHD symptoms leading to problems following or starting routines.

Gold Star is a habit-building and time management solution designed to help users organize their schedules, break down tasks, and build sustainable routines.

We are focused on adaptability, flexible scheduling, and customizing the experience for every user.

Appendix



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